

Excel Practice Pack

Questions, hints and answers for the workbook **Excel_Practice_Dataset.xlsx**. Covers every topic in your course (Sessions 1-21).

How to use this pack

1. Open the workbook. Each sheet is a small business dataset (2024 sales, products, employees, customers, targets...). Work on a **copy** of the sheet.
2. Read the question in Part A and try it **without** looking at the hint.
3. Stuck after ~5 minutes? Read the hint (it names the function or the menu path, never the final answer).
4. Compare your result with Part B. If it differs, check the formula with F2 / Evaluate Formula before peeking at the answer's formula.
5. Put a tick next to each question ID (e.g. S9.3) you got right first time - repeat the rest after a day.

Sheets in the workbook

Sheet	What it holds	Sessions
S1_DataTypes	30 records with wrong data types (text-numbers, text-dates, mixed booleans, errors)	1, 16B
Sales_Data	800 orders in 2024 (Region, SalesRep, ProductID, Quantity, Discount, Channel, Delivered, SalesAmount)	2-4, 7-21
Products	20 products: Category, UnitPrice, UnitCost	5, 9-11, 13-14, 18-21
Employees	30 employees: messy names, codes, dates, marks. L2 = fixed as-of date (31-Dec-2025)	5-8, 12, 21
Bonus_History	Several bonus payments per employee (date order)	10
Lookup_Tables	Grade slabs, commission slabs, horizontal table, dropdown lists, holidays	2, 7-10
Targets	Monthly targets per region (2024)	11, 12, 21
Customers_Messy	Messy names/phones/addresses/emails, blank rows, duplicates	3, 6, 12, 16B, 17
Sales_Extra	50 orders Jan-Mar 2025, different column names/order, blank discounts	18, 21

*Good to know: the data is synthetic and currency-neutral. The 6 orders with ProductID **P099** are missing from Products on purpose (to practise #N/A handling and merge nulls). I used a fixed date (Employees!L2) instead of TODAY() so your answers never change.*

Suggested 5-week plan (about 60-90 min per block)

Week	Focus	Sessions	How to practise
1	Foundations: types, references, validation, sort/filter, text, dates, logic	1-8	Do all questions once; redo the formula-writing ones from a blank cell next day.
2	Lookups + conditional formatting	9-12	Write the same lookup 3 ways (VLOOKUP, XLOOKUP, INDEX/MATCH) using the 'Which function when' page.
3	Pivot tables, pivot charts, slicers	13-16A	Build every pivot from scratch, then compare numbers with the answer tables.
4	Cleaning tools + Power Query	16B-18	Repeat 16B and 17 in Power Query and compare row counts.
5	Dashboard design + capstone	19, 21	Time yourself: build the full dashboard in 90 minutes, then refine.

Which function when? (quick reference)

I want to...	Use	Watch out for
Take the first / last N characters	LEFT, RIGHT	Fixed length only - use FIND for variable length
Take characters from the middle	MID(text,start,n)	Combine with FIND to locate a delimiter
Find the position of a character	FIND (case-sens.), SEARCH (wildcards)	Errors with #VALUE! if not found
Count characters	LEN	Spaces count too
Remove extra spaces / hidden characters	TRIM, CLEAN, SUBSTITUTE(x,UNICHAR(160)," ")	TRIM does not remove UNICHAR(160) or line breaks
Replace text	SUBSTITUTE(text,old,new)	Case-sensitive; REPLACE works by position
Fix capitalisation	UPPER, LOWER, PROPER	PROPER turns McDonald into Mcdonald
Format a number/date as text	TEXT(x,"mmm-yyyy")	Result is text, not a number
Repeat a text (mini bars)	REPT(" ",n)	n must be a whole number
Today's date / date + time	TODAY(), NOW()	Volatile - changes every recalculation
Year / month / day of a date	YEAR, MONTH, DAY	Date must be a real date, not text
Build a date	DATE(y,m,d)	Month 14 or day 0 rolls over
Years / months / days between dates	DATEDIF(start,end,"Y"/"M"/"D")	Hidden function; start must be <= end
Last day of a month	EOMONTH(date,months)	0 = same month, 1 = next month
Working days between dates	NETWORKDAYS(start,end,holidays)	Includes both start and end dates
One condition -> two outcomes	IF	Nest carefully; highest threshold first
Several conditions at once	AND, OR, XOR	XOR = exactly one is TRUE
Many tiers without nesting	IFS	Finish with TRUE,"default"
Look up vertically (exact)	VLOOKUP(x,table,col,FALSE)	Return column must be right of the key; hard-coded col number
Look up in a band (slabs)	VLOOKUP(x,table,col,TRUE) or XLOOKUP(...,-1)	Sort ascending for TRUE
Look up horizontally	HLOOKUP, XLOOKUP	Row index counts rows in the chosen table
Best all-round lookup (365)	XLOOKUP(x,lookup,return,default,match,search)	Search -1 = last match; match 2 = wildcards
Look left / flexible / any version	INDEX(return,MATCH(x,lookup,0))	Both ranges same size; 0 = exact
Two-way table lookup	INDEX(grid,MATCH(row),MATCH(col))	Two MATCH functions
Hide errors	IFERROR, IFNA	Do not hide real problems - check the cause
Count / sum with condition (bonus)	COUNTIFS, SUMIFS	Criteria pairs; dates need ">="&DATE(...)
Count many things without helpers	SUMPRODUCT(--(condition))	Wrap text-numbers with --
Spot duplicates / top N / scales	Conditional Formatting	Formula rules need \$ on the column
Summarise big data fast	PivotTable + Slicers/Timeline	Refresh after data changes; group dates
Split text into columns	Text to Columns, Flash Fill (Ctrl+E)	Flash Fill needs consistent patterns
Combine / join tables reliably	Power Query: Merge (join) and Append	Column names must match for Append

Lookup decision in 10 seconds: need to look left, or worried about inserted columns -> INDEX/MATCH or XLOOKUP. Need last match or a default value -> XLOOKUP. Two dimensions (row + column) -> INDEX +

MATCH + MATCH. Slabs/bands -> approximate match on a table sorted ascending. Older Excel or simple exact lookup -> VLOOKUP with FALSE.

PART A - Questions & Hints

Answers are in Part B (Answer Key) at the end - try first!

Session 1 - Interface, Layout & Data Types

Sheet: S1_DataTypes (A1:E31). Goal: spot wrong data types and fix them.

- S1.1** In S1_DataTypes, how many cells in **Amount (B2:B31)** are real numbers and how many are text that only looks like a number?
Hint: COUNT counts only true numbers, COUNTA counts everything non-blank. Text can be counted with SUMPRODUCT + ISTEEXT. Left-aligned + green triangle = text.
- S1.2** What does =SUM(B2:B31) return as the data stands? Then convert the text-numbers to real numbers and give the corrected total. Which trick do you prefer (VALUE, multiply by 1, Text to Columns, Convert to Number)?
Hint: SUM ignores text. To get the corrected total without changing data, try =SUMPRODUCT(--B2:B31) or a helper column =VALUE(B2).
- S1.3** In JoinDate (C2:C31) how many entries are real dates and how many are text? After converting all of them to real dates, what is the **earliest** join date?
Hint: =COUNT(C2:C31) counts real dates (dates are numbers). For text dates use DATEVALUE(), Text to Columns (Date > YMD) or --. Format the result as dd-mmm-yyyy.
- S1.4** The **Active (D2:D31)** column mixes real booleans (TRUE/FALSE) with text ("Yes"/"No"). How many records are active?
Hint: COUNTIF(range,TRUE) counts real TRUE only; COUNTIF(range,"Yes") counts the text. Add them.
- S1.5** In **Score (E2:E31)** count (a) real error cells (#N/A, #VALUE!) and (b) the text entries "N/A". Then calculate the average of the valid numeric scores ignoring both.
Hint: ISERROR/ISERR inside SUMPRODUCT counts errors. AVERAGE stops at an error; AGGREGATE(1,6,range) ignores errors and text is ignored by AVERAGE anyway.
- S1.6** What number does =TYPE(x) return for: (a) a number, (b) text, (c) TRUE, (d) an error value? Use it on B3, D2 and E8 to confirm which are which on your sheet.
Hint: TYPE codes are 1, 2, 4, 16 (and 64 for arrays).
- S1.7** R1C1 mode: in cell D5 the formula is =RC[-1]*R2C3. Rewrite it in normal A1 style. Where do you switch reference styles?
Hint: RC[-1] = same row, one column to the left. R2C3 = row 2, column 3 (absolute).

Session 2 - Cell Referencing, Named Ranges, Data Entry Rules

Sheets: Products, Sales_Data, Lookup_Tables. Practise on a copy of the sheet.

- S2.1** Create a **static named range** PriceList for Products!A2:E21 (Name Box or Formulas > Define Name). Using it, find the UnitPrice of ProductID P005.
Hint: =VLOOKUP("P005",PriceList,4,FALSE). UnitPrice is the 4th column.
- S2.2** Create a **dynamic named range** SalesAmt for the SalesAmount column that grows automatically, then calculate =SUM(SalesAmt). Add a new row 802 with SalesAmount = 1000 and check the total increases by 1000.
Hint: Name refers to: =Sales_Data!\$J\$2:INDEX(Sales_Data!\$J:\$J,COUNTA(Sales_Data!\$J:\$J)) (OFFSET also works but is volatile).
- S2.3** Build a **dropdown for Product Category**: in a blank cell, apply Data Validation (List) using the Category list in Lookup_Tables!J2:J6. Make it show an error stop message "Pick a category from the list".
Hint: Data > Data Validation > Allow: List > Source: =Lookup_Tables!\$J\$2:\$J\$6 (or name the range CategoryList and use =CategoryList). Error Alert style = Stop.
- S2.4** On Sales_Data!F2:F801 apply validation: **Quantity must be a whole number between 1 and 50**. Which value would be rejected: 0, 12, 12.5, 50?
Hint: Allow: Whole number > between 1 and 50.

S2.5 Apply date validation on B2:B801 so **OrderDate must fall in 2024**. Which of these are rejected: 31-Dec-2023, 01-Jan-2024, 29-Feb-2024, 01-Jan-2025?

Hint: Allow: Date > between 01-01-2024 and 31-12-2024 (or use DATE(2024,1,1) and DATE(2024,12,31) inside the boxes).

S2.6 Custom rule: in A2:A1000 allow only OrderIDs that start with "ORD-" AND are not already used. Write the validation formula.

Hint: Allow: Custom. Combine LEFT and COUNTIF with AND. Formula is written for the first selected cell (A2) with a *mixed* reference for the range.

S2.7 Mixed references: on a blank sheet type the unit prices of P001..P005 across B1:F1 and quantities 1, 5, 10, 20, 50 down A2:A6. Write ONE formula in B2 that can be copied to the whole B2:F6 grid. What is the value in E5 and which reference style did you use?

Hint: Lock the column for the quantities and the row for the prices: \$A2 and B\$1.

Session 3 - Sorting, Basic & Advanced Filtering

Sheets: Sales_Data, Customers_Messy.

S3.1 Multi-level sort Sales_Data by Region (A-Z), then ProductID (A-Z), then SalesAmount (largest to smallest). What are the Region, ProductID and SalesAmount in the first data row?

Hint: Data > Sort > Add Level twice. Keep 'My data has headers' ticked.

S3.2 Use AutoFilter: Region = East, Channel = Online, Delivered = FALSE. How many orders are left?

Hint: Look at the status bar ('x of 800 records found') or use =SUBTOTAL(103,A2:A801) under the filtered data.

S3.3 With AutoFilter > Number Filters > **Top 10**, show the top 10 items in SalesAmount. What is the value of the 10th largest order?

Hint: Cross-check with =LARGE(J2:J801,10).

S3.4 Advanced Filter (criteria range): show orders where SalesAmount > 5000 AND Region = East. How many rows?

Hint: Copy headers SalesAmount and Region to a blank area. Under them, in the SAME row, type >5000 and East. Data > Advanced > List range A1:J801, Criteria range = your 2x2 area.

S3.5 In Customers_Messy, how many **completely blank rows** are inside the data? After deleting them, how many data rows remain (excluding the header)?

Hint: Select column A > F5 > Special > Blanks > Home > Delete > Delete Sheet Rows. Careful: only safe if the row is blank across ALL columns.

Session 4 - Advanced Filtering & Custom Views

Sheet: Sales_Data. Use a criteria area to the right of the data (e.g. L1:P4).

S4.1 OR condition: show orders where (Region = East AND SalesAmount > 5000) OR (Region = West AND SalesAmount > 8000). How many rows?

Hint: Criteria range with headers Region and SalesAmount. Row 2: East | >5000. Row 3: West | >8000. Different rows = OR, same row = AND.

S4.2 Copy to another location: filter Region = North, Delivered = FALSE, Discount >= 10% and copy results to a new area (e.g. column R). How many result rows (excluding header)?

Hint: Advanced > 'Copy to another location'. Criteria: Region North, Delivered FALSE, Discount >=0.1 all in one row. 'Copy to' = a single header cell like \$R\$1.

S4.3 Using Advanced Filter with **Unique records only**, list the distinct sales reps who sold in the South region.

Hint: Criteria Region = South. Copy to another location with only the SalesRep header in 'Copy to', tick Unique records only.

S4.4 Wildcard criteria: filter SalesRep names beginning with the letter A. How many orders?

Hint: Criteria cell under SalesRep header: A*.

S4.5 Same column twice (AND): show orders with SalesAmount between 2000 and 4000 inclusive.

How many?

Hint: Put the SalesAmount header twice side by side in the criteria range: `>=2000` and `<=4000` in the same row.

S4.6 Custom Views: save one view named "East_Big" (filter Region=East, Sales>5000, columns H:I hidden) and one named "All" (no filters, everything visible). How do you switch, and why might the command be greyed out?

Hint: View > Custom Views > Add. Think about what Excel objects block this feature.

Session 5 - Text Functions (Part 1)

Sheets: Employees (Code column C), Products, Sales_Data.

S5.1 Employees!C has codes like `TX-2345` (state - number). (a) How many employees are in state TX? (b) What is the sum of all the 4-digit numbers?

Hint: `LEFT(C2,2)` gives the state; `RIGHT(C2,4)` gives the number as text - convert with `--`. Wrap in `SUMPRODUCT` to avoid helper columns. Excel comparisons ignore case.

S5.2 Split Products!B (like `Tablet-Samsung`) into Category and Brand with formulas (no Text to Columns). What are the category, brand and length of the brand for ProductID **P010**?

Hint: `=LEFT(B2,FIND("-",B2)-1)` and `=MID(B2,FIND("-",B2)+1,LEN(B2))`. `LEFT` with a fixed number fails because category lengths differ.

S5.3 Which Product name in Products!B2:B21 is the **longest** and how many characters does it have?

Hint: `LEN` gives the length; `=MAX(...)` over an array of `LEN` values (helper column is fine).

S5.4 Show the OrderDate of Sales_Data row 2 as text in the format `Monday, 05 Feb 2024` (your date will differ). What does `=TEXT(B2,"dddd, dd mmm yyyy")` return?

Hint: `TEXT` converts a number/date to text using a format code: `dddd` = full weekday, `mmm` = short month.

S5.5 Make an in-cell bar chart: in column K use `REPT` so that one `|` represents 500 of SalesAmount (rounded). How many bars are shown for **row 3**?

Hint: `=REPT("|",ROUND(J3/500,0))`

S5.6 Extract the numeric part of OrderID as a real number (`ORD-1001` -> 1001). What do you get for the last order (row 801) and what is `=LEN(A801)`?

Hint: `=VALUE(MID(A801,5,4))` or `--RIGHT(A801,4)`.

S5.7 Demo drill: for EmpID **E010** build the label `STATE-LAST2` in capitals, e.g. `TX-45`, from the Code in column C. What is it?

Hint: `=UPPER(LEFT(C11,2))&"-"&RIGHT(C11,2)`. Row 11 is E010.

Session 6 - Text Functions (Part 2: Cleaning)

Sheets: Employees, Customers_Messy. Hidden traps: non-breaking spaces (UNICHAR(160)) and line breaks (CHAR(10)).

S6.1 Clean Employees!B (FullName_Raw) into proper names like `Rahul Shah`. What is the cleaned name for **E005** (raw value: see cell B6)?

Hint: `TRIM` removes extra spaces but NOT non-breaking spaces or line breaks (`CHAR(160)` also works on Windows, `UNICHAR(160)` works everywhere). Use `=PROPER(TRIM(CLEAN(SUBSTITUTE(B2,UNICHAR(160),""))))`.

S6.2 How many of the 30 raw names are NOT already in their clean form, and how many characters were removed in total (sum of `LEN` raw - sum of `LEN` clean)?

Hint: Compare with `EXACT` (case-sensitive) because `=` ignores case:
`=SUMPRODUCT(--NOT(EXACT(B2:B31,D2:D31)))` where D = cleaned. Total removed =
`=SUMPRODUCT(LEN(B2:B31))-SUMPRODUCT(LEN(D2:D31))`.

S6.3 Some Codes in Employees!C are lowercase. How many codes are not fully uppercase? Standardise them with `UPPER`.

Hint: `EXACT(C2,UPPER(C2))` is `FALSE` for lowercase codes.

- S6.4** Customers_Messy!C has phones like (555) 123-4567, 555.123.4567, 555-123-4567. Convert every phone to 10 digits only. What is the clean phone of customer **C001**?
Hint: Nest SUBSTITUTE to remove (,), space, - and ..
- S6.5** After =LOWER(TRIM(E2)), how many rows in Customers_Messy!E2:E49 have an email ending in @gmail.com (duplicates still included, blank rows ignored)?
Hint: =SUMPRODUCT(--(RIGHT(LOWER(TRIM(E2:E49)),10)="@gmail.com")). "@gmail.com" has 10 characters.
- S6.6** From the cleaned name, split first and last name with LEFT/RIGHT + FIND. What are the first name and the length of the last name for **E008**?
Hint: =LEFT(x,FIND(" ",x)-1) and =MID(x,FIND(" ",x)+1,99).
- S6.7** Standardise the address of customer **C005**: TRIM extra spaces, remove any space before a comma, then use SUBSTITUTE to change " Street" to " St" and " Avenue" to " Ave". What is the final text?
Hint: =SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(TRIM(D2)," ","")," Street"," St")," Avenue"," Ave").
 SUBSTITUTE is case-sensitive.

Session 7 - Date & Time Functions

Sheets: Employees (fixed as-of date in L2 = 31-Dec-2025 - use \$L\$2 instead of TODAY() so answers stay the same), Sales_Data, Lookup_Tables.

- S7.1** Calculate each employee's **age** as of Employees!\$L\$2. What is E001's age, and how many employees are older than 40?
Hint: =DATEDIF(E2,\$L\$2,"Y") in a helper column, then =COUNTIF(K2:K31,">40"). (DATEDIF is hidden from the function list - type it fully.)
- S7.2** Calculate **tenure in full years** from HireDate to EndDate (or to \$L\$2 if EndDate is blank). Who has the longest tenure and how many years?
Hint: =DATEDIF(F2,IF(G2="", \$L\$2, G2), "Y"). Use cleaned names to report.
- S7.3 NETWORKDAYS:** for employee E002 (has an EndDate) how many working days (Mon-Fri) between HireDate and EndDate, both inclusive?
Hint: =NETWORKDAYS(F2,G2) - use the correct row for that employee.
- S7.4** Working days in **January 2025** (01-Jan-2025 to 31-Jan-2025) excluding the holidays listed in Lookup_Tables!H10:H13?
Hint: =NETWORKDAYS(DATE(2025,1,1),DATE(2025,1,31),Lookup_Tables!H10:H13). One of the holidays falls on a weekend - is it double-counted?
- S7.5** For the order in Sales_Data row 2 (date 02-Jan-2024): (a) last day of that month, (b) last day of the NEXT month (a payment due date).
Hint: =EOMONTH(B2,0) and =EOMONTH(B2,1). Format results as dates.
- S7.6** Using MONTH / DAY on OrderDate: (a) how many orders were placed in March 2024, (b) in Q4 (Oct-Dec), (c) on day 25 or later of any month?
Hint: =SUMPRODUCT(--(MONTH(B2:B801)=3)), =SUMPRODUCT(--(MONTH(B2:B801)>=10)), =SUMPRODUCT(--(DAY(B2:B801)>=25)).
- S7.7** Predict the results without a computer: =DATE(2024,14,1) and =DATE(2024,3,0). What happens when month or day is out of range?
Hint: DATE rolls over: month 14 = 2 months into next year; day 0 = the day before the 1st.
- S7.8** Predict =DATEDIF(DATE(2024,3,15),DATE(2025,12,31),"Y"), "M", "YM" and "MD". Also: what is the difference between TODAY() and NOW(), and how do you type a static date?
Hint: Y = whole years, M = whole months, YM = months ignoring years, MD = days ignoring months and years. TODAY has no time part.

Session 8 - Logical Functions

Sheets: Employees (Marks H, Attendance I, Certified J), Sales_Data.

- S8.1 Nested IF grading:** A >= 90, B >= 75, C >= 60, D >= 50, otherwise F. How many employees get each grade?
Hint: =IF(H2>=90,"A",IF(H2>=75,"B",IF(H2>=60,"C",IF(H2>=50,"D","F")))). Put the highest condition first. Count with COUNTIF on the grade column.
- S8.2** Write the same grading with **IFS** (Excel 2019+). What grade does E010 get and what is the last condition you need for the 'otherwise' case?
Hint: IFS has no 'else' argument - finish with TRUE, "F".
- S8.3 AND:** an employee is *bonus-eligible* if Marks >= 75 AND Attendance >= 90. How many are eligible?
Hint: =IF(AND(H2>=75,I2>=90),"Eligible","No"), then COUNTIF.
- S8.4 OR:** an employee 'needs training' if Marks < 50 OR Attendance < 75. How many?
Hint: =OR(H2<50,I2<75) returns TRUE/FALSE; count TRUEs with =COUNTIF(range,TRUE).
- S8.5 XOR:** flag employees where EXACTLY ONE of these is true: Marks >= 75, Certified = "Yes". How many TRUE results?
Hint: =XOR(H2>=75,J2="Yes"). XOR with two arguments is TRUE only when they differ.
- S8.6 Order tiers** on SalesAmount: >= 10000 Platinum, >= 5000 Gold, >= 2000 Silver, else Bronze. How many orders per tier?
Hint: Nested IF or IFS in a helper column K, then COUNTIF.
- S8.7 IF + AND:** flag an order 'Priority follow-up' when Delivered = FALSE and SalesAmount > 5000. How many flagged orders?
Hint: =IF(AND(I2=FALSE,J2>5000),"Priority",""); count with =COUNTIF(K2:K801,"Priority").

Session 9 - Lookup Functions (VLOOKUP / HLOOKUP)

Sheets: Sales_Data, Products, Lookup_Tables. Add helper columns to the right of Sales_Data (K onward).

- S9.1** Add **UnitPrice** in K2 using VLOOKUP on Products. What is the price for row 2 (ProductID P013)? How many rows return #N/A?
Hint: =VLOOKUP(E2,Products!\$A\$2:\$E\$21,4,FALSE). Lock the table with \$ and use FALSE for exact match. Count errors with =SUMPRODUCT(--ISNA(K2:K801)).
- S9.2** Check the data: in L2 test =ROUND(K2*F2*(1-G2),2)=J2. How many of the 794 valid rows are FALSE (mismatches)? Wrap in IFERROR so the 6 bad rows show "Missing product".
Hint: =IFERROR(ROUND(K2*F2*(1-G2),2)=J2,"Missing product"); then COUNTIF(L2:L801,FALSE).
- S9.3 Approximate match:** get the commission rate for every order from Lookup_Tables!D2:E5 (0 -> 1%, 2000 -> 2%, 5000 -> 3.5%, 10000 -> 5%). Then Commission = SalesAmount x rate. What is the total commission?
Hint: =VLOOKUP(J2,Lookup_Tables!\$D\$2:\$E\$5,2,TRUE). The first column MUST be sorted ascending for approximate match.
- S9.4** Use approximate-match VLOOKUP with Lookup_Tables!A2:B6 to grade Employees!H (Marks). Do your grade counts match your Session 8 answers?
Hint: =VLOOKUP(H2,Lookup_Tables!\$A\$2:\$B\$6,2,TRUE).
- S9.5 Limitation:** find the ProductID for the product code "Laptop-Dell" (a lookup to the LEFT). Can VLOOKUP do it directly? What is the answer, and which function would you switch to?
Hint: VLOOKUP always returns from columns to the RIGHT of the lookup column.
- S9.6 HLOOKUP:** using the horizontal table Lookup_Tables!A9:F11, return (a) ReturnDays for Tablet, (b) WarrantyYears for Laptop.
Hint: =HLOOKUP("Tablet",Lookup_Tables!\$B\$9:\$F\$11,3,FALSE). Row index counts rows of the table array you selected.
- S9.7 Column-index trap:** you insert a new column between Category and UnitPrice in Products. What happens to VLOOKUP(E2,Products!A:E,4,FALSE)? How do you make it safe?
Hint: The 4 is hard-coded; the range expands to A:F after insertion. What is now in column 4?

Session 10 - Modern Lookup (XLOOKUP)

Sheets: Sales_Data, Products, Bonus_History, Lookup_Tables. XLOOKUP needs Excel 2021 / 365.

- S10.1** Rewrite the Session 9 price lookup with **XLOOKUP** and a friendly default: "Not found". How many rows show "Not found"?
Hint: `=XLOOKUP(E2,Products!$A$2:$A$21,Products!$D$2:$D$21,"Not found")`. Exact match is the default - no FALSE needed.
- S10.2 Reverse lookup:** return the ProductID for the product "Audio-Sony" and the ProductID for "Wearable-Garmin".
Hint: `Lookup array = Products!B2:B21, return array = Products!A2:A21`.
- S10.3 Last bonus (search from bottom):** Bonus_History is in date order. What is the LAST bonus amount and date for employee **E002**, and the FIRST one? What should `=XLOOKUP("E003",...)` show if you set the default to "No bonus"?
Hint: `=XLOOKUP("E002",Bonus_History!A:A,Bonus_History!C:C,"No bonus",0,-1)` - the last argument -1 searches last-to-first. Default (1) gives the first match.
- S10.4** Sales_Data is in date order. What is the SalesAmount and OrderID of the **most recent order** by Amit Verma?
Hint: Search Sales_Data!D2:D801 for the name with search_mode = -1, return column J (and A).
- S10.5 Approximate match with XLOOKUP:** the commission rate (Lookup_Tables!D2:E5) for orders of 4999.99, 5000 and 10000? Which match_mode do you use?
Hint: match_mode -1 = exact or next smaller item.
`=XLOOKUP(x,Lookup_Tables!D2:D5,Lookup_Tables!E2:E5,, -1)`.
- S10.6 Return several columns at once:** what does `=XLOOKUP("P010",Products!A2:A21,Products!B2:E21)` spill into the sheet?
Hint: Return array may be several columns wide.
- S10.7 Wildcards:** with match_mode 2, what does `=XLOOKUP("*Dell",Products!B2:B21,Products!D2:D21,"None",2)` return, and what does `=XLOOKUP("Wearable-*",Products!B2:B21,Products!D2:D21,"None",2)` return?
Hint: * matches any characters; the first match from the top is returned.

Session 11 - INDEX-MATCH & MATCH-MATCH

Sheets: Products, Targets, Lookup_Tables, Sales_Data.

- S11.1** Use **INDEX/MATCH** to return the UnitPrice for ProductID P012.
Hint: `=INDEX(Products!$D$2:$D$21,MATCH("P012",Products!$A$2:$A$21,0))`. The 0 = exact match.
- S11.2** Which product has the **highest** UnitPrice and which has the **lowest**? Return the Product names with INDEX/MATCH + MAX/MIN.
Hint: `=INDEX(Products!B2:B21,MATCH(MAX(Products!D2:D21),Products!D2:D21,0))`
- S11.3 Two-way lookup (MATCH-MATCH):** the monthly target for (a) West / Mar, (b) South / Nov, (c) East / Aug.
Hint:
`=INDEX(Targets!$B$2:$M$5,MATCH("West",Targets!$A$2:$A$5,0),MATCH("Mar",Targets!$B$1:$M$1,0))`. First MATCH = row number, second = column number.
- S11.4** Make the two-way lookup **dynamic**: put a Region dropdown in a cell and a Month dropdown in another and reference those cells inside the MATCH functions. Test with North / Jun and compare with actual June sales in North (SUMIFS on Sales_Data). By how much did North miss or beat its target?
Hint: Actual: `=SUMIFS(Sales_Data!J:J,Sales_Data!C:C,"North",Sales_Data!B:B,">="&DATE(2024,6,1),Sales_Data!B:B,"<="&DATE(2024,6,30))`.
- S11.5** What do these return? (a) `=MATCH("P020",Products!A2:A21,0)` (b) `=MATCH("P099",Products!A2:A21,0)` (c) `=MATCH(7500,Lookup_Tables!D2:D5,1)`
Hint: Type 0 = exact, 1 = largest value <= lookup (sorted ascending).

S11.6 Why is INDEX/MATCH safer than VLOOKUP when columns are inserted? Write the formula that returns UnitCost for ProductID P007 and give the answer.

Hint: INDEX points at the return column itself; nothing is counted by position.

Session 12 - Conditional Formatting

Sheets: Customers_Messy, Sales_Data, Employees, Targets. Counts below are what you should see highlighted.

S12.1 Highlight duplicates in Customers_Messy!E2:E49 (Email). How many cells get highlighted? What does that tell you about the duplicate rows?

Hint: Home > Conditional Formatting > Highlight Cells Rules > Duplicate Values. Duplicates are highlighted in BOTH the original and the copy.

S12.2 Apply a 3-colour scale (red low, yellow mid, green high) to SalesAmount J2:J801. What are the values at the two ends (lowest / highest) that get pure red and pure green?

Hint: Use =MIN(J2:J801) and =MAX(J2:J801) to check the extremes.

S12.3 Icon set (traffic lights) on Employees!I2:I31 Attendance: green >= 90, yellow >= 75, red < 75 (set type = Number, not Percent). How many green / yellow / red?

Hint: Manage Rules > Edit > Icon Style 3 Traffic Lights > change both thresholds to Type: Number and enter 90 and 75. Untick the wrong default percentages.

S12.4 Top/Bottom N: highlight the Top 10 values in SalesAmount. What is the lowest value that is still highlighted and how many cells end up highlighted? Also list the 5 smallest amounts (Bottom 5).

Hint: Top/Bottom Rules > Top 10 Items. Ties at the 10th value can highlight more than 10 cells.

S12.5 Revenue vs target: build a Region x Month summary of actual 2024 sales (PivotTable or SUMIFS), compare each cell with Targets!B2:M5 and colour it green if actual > target, red if actual < target. How many of the 48 region-months are green, per region?

Hint: Use a formula-based rule such as =B2>Targets!B2 on your summary grid (cell reference to another sheet works in Excel 2010+). Or put the targets grid next to it.

S12.6 Formula rule for whole rows: shade the entire Sales_Data row (A:J) when Delivered = FALSE AND SalesAmount > 5000. How many rows are shaded and which formula do you use?

Hint: Use 'Use a formula to determine which cells to format'. Lock the column, not the row: \$I2.

Session 13 - Pivot Tables (Basics)

Source: Sales_Data (A1:J801). For Category you first need a helper column (lookup from Products; use "Unknown" for the 6 missing products).

S13.1 Insert a PivotTable: Sum of SalesAmount by Region. What are the four totals and the grand total?

Hint: Insert > PivotTable > Rows: Region, Values: SalesAmount (Sum).

S13.2 Put Region and SalesRep in Rows (in that order). Which are the **top 3 reps in each region** by total sales?

Hint: Right-click a rep total > Sort > Largest to Smallest. Within each region reps are sorted separately.

S13.3 Add a helper column Category to Sales_Data (K). Then build a PivotTable: Category in Rows, Region in Columns, Sum of SalesAmount. Fill in the totals (rounded to whole numbers).

Hint: =IFERROR(VLOOKUP(E2,Products!\$A\$2:\$C\$21,3,FALSE),"Unknown") then refresh the pivot source range to A1:K801.

S13.4 Show Channel in Rows and Values: Count of OrderID, Average of SalesAmount and Sum of SalesAmount (rename the headings). Fill in the table.

Hint: Drag SalesAmount twice into Values and change Value Field Settings > Summarize by (Average / Sum). Count OrderID as a third value.

S13.5 Put Delivered in Filters and select FALSE. How many undelivered orders and what total value?

Hint: Count of OrderID and Sum of SalesAmount with the report filter Delivered = FALSE.

S13.6 Change the Region totals to **% of Grand Total** (Value Field Settings > Show Values As). What percentage does each region contribute (1 decimal)?

Hint: Show Values As > % of Grand Total.

Session 14 - Pivot Tables (Advanced)

Source: Sales_Data with helper columns Category (K), UnitCost (L), Profit (M).

S14.1 Put OrderDate in Rows and **Group by Months and Quarters**. Give total sales for Q1-Q4 and name the best single month with its total.

Hint: Right-click a date > Group > select Months + Quarters (+Years). In Excel 365 grouping may be automatic.

S14.2 Build Region (rows) x Quarter (columns). Fill in the table.

Hint: Same grouping as above, with Region dragged into Rows and Quarters into Columns.

S14.3 **Calculated field:** add helper column Profit = SalesAmount - Quantity x UnitCost (UnitCost from Products; leave blank/0 for unknown products and filter Category <> Unknown). Then insert a calculated field **Margin = Profit / SalesAmount**. Fill in the table for the 5 categories.

Hint: PivotTable Analyze > Fields, Items & Sets > Calculated Field. Formula: = Profit / SalesAmount. Calculated fields work on SUMS, not row by row.

S14.4 Insert **slicers** for Region and Channel. Select Region = East and Channel = Online. What are the order count and total sales?

Hint: PivotTable Analyze > Insert Slicer. Ctrl-click multiple items. Show Count of OrderID and Sum of SalesAmount.

S14.5 **Drill-down:** double-click the West total in a Region pivot. How many records open on the new sheet? And for the Grand Total?

Hint: Double-clicking a value cell in a pivot extracts the underlying rows.

S14.6 Apply a **Value Filter (Top 10 > Top 3 by Sum of SalesAmount)** to SalesRep. Who are the top 3 reps overall and their totals?

Hint: Row Labels dropdown > Value Filters > Top 10 > show Top 3 Items by SalesAmount.

Session 15-16A - PivotCharts, Slicers, Timelines & Dashboards

Source: Sales_Data (with Category helper column). Numbers below let you verify what your chart shows.

S15-16A.1 Create a **PivotChart (Line)** of monthly Sales (Jan-Dec). What are the 12 values plotted?

Hint: PivotTable with OrderDate grouped by Month (only Months), Values = Sum of SalesAmount, then Analyze > PivotChart > Line.

S15-16A.2 Create a **clustered column** PivotChart of sales by Category (include Unknown). Read the bar heights.

Hint: Rows: Category, Values: Sum of SalesAmount. Sort largest to smallest for readability.

S15-16A.3 Create a **Pie / Donut** chart of Channel share. What percentage is each slice? Which chart type would you choose for 12 categories instead and why?

Hint: Values > Show Values As > % of Grand Total. Pie is only readable with a few slices.

S15-16A.4 Insert a **Timeline** slicer on OrderDate and select Jul-Sep 2024. What is total sales? Then also click Region = South on a Region slicer - what is the total now?

Hint: PivotTable Analyze > Insert Timeline > drag across Jul, Aug, Sep. Slicers combine with AND logic.

S15-16A.5 Make ONE Region slicer control TWO PivotTables/PivotCharts (e.g. monthly line chart and channel donut). How? With Region = West selected, what total should both show?

Hint: Right-click slicer > Report Connections > tick both pivots.

S15-16A.6 **Design principles quiz:** pick the best chart: (a) trend over 12 months, (b) compare 5 categories, (c) share of 3 channels, (d) part-to-whole over time. Name two formatting rules for a clean dashboard.

Hint: Think about what the eye must compare: position along a scale, angles, or area.

Session 16B - Data Cleaning Tools & Error Handling

Sheets: Customers_Messy, S1_DataTypes. Work on a copy.

S16B.1 Delete the blank rows in Customers_Messy, then use **Remove Duplicates** on ALL columns. How many rows are removed and how many unique customers remain?

Hint: Data > Remove Duplicates > select all columns > OK. Excel reports the number removed.

S16B.2 Instead run Remove Duplicates on the **Email** column only (on the raw data). How many rows are removed? Why is Email a better key than the Name column here?

Hint: Names are messy (case, spaces) so identical people look different; emails were copied exactly.

S16B.3 Use **Text to Columns** (Delimited, comma) on the Address column to split Street, City and 'State ZIP'. Then TRIM everything. How many DISTINCT cities do the 40 unique customers live in and how many are in Texas (TX)? Tip: split 'State ZIP' with a space delimiter.

Hint: Some addresses have no space after commas or extra spaces before/after them - TRIM fixes it. Text to Columns > tick 'Treat consecutive delimiters as one' for the space split.

S16B.4 Use **Flash Fill** (Ctrl+E) to extract the FIRST name from the cleaned name column. What first name appears for customer **C010**?

Hint: Type the first two first-names manually in the next column, select the range and press Ctrl+E. Flash Fill needs clean, consistent examples.

S16B.5 Find & Replace with wildcards: select Notes (F2:F49), replace **VIP-*** with **VIP**. How many replacements does Excel report?

Hint: Ctrl+H, Find what: **VIP-***, Replace with: **VIP**, Replace All. Select the column first so nothing else changes. ? = any one character, ~* = a literal star.

S16B.6 In S1_DataTypes select B2:B31 and use the green triangle > **Convert to Number**. What does **=COUNT(B2:B31)** show before and after? Then use **IFERROR** to replace the error cells of Score with 0: what is **=AVERAGE(...)** of the helper column?

Hint: Helper: **=IFERROR(E2,0)**. Text "N/A" is NOT an error, so AVERAGE ignores it. Formulas > Error Checking / Evaluate Formula help you trace errors.

Session 17 - Power Query (Basics)

Data > Get Data > From Table/Range on Sales_Data or Customers_Messy. Use the Applied Steps pane.

S17.1 Load Sales_Data into Power Query. What data types should the columns be (OrderDate, Discount, Delivered, SalesAmount) and how many rows load?

Hint: Power Query auto-detects types in a 'Changed Type' step. Check each column header icon.

S17.2 Remove all columns except OrderID, OrderDate, Region, ProductID, Quantity, SalesAmount. How many columns remain and what do you call the step?

Hint: Select the columns you want > right-click > Remove Other Columns.

S17.3 From the original 10 columns, filter Delivered = TRUE. How many rows remain?

Hint: Filter icon on Delivered > uncheck FALSE.

S17.4 Add a **Custom Column** YearMonth with **Date.ToText([OrderDate], "yyyy-MM")**. What is the value in the first row?

Hint: Add Column > Custom Column. Watch the capitalisation: Power Query is case-sensitive.

S17.5 Use **Group By**: Region, Sum of SalesAmount. Do the values match your Session 13 pivot?

Hint: Transform > Group By.

S17.6 Extract the number from OrderID (ORD-1001 -> 1001) with Extract > Text After Delimiter ("-"), change type to Whole Number. What are the min and max?

Hint: Transform > Extract > Text After Delimiter; then change type; then Transform > Statistics if you want min/max.

S17.7 Load Customers_Messy into Power Query: Remove Blank Rows, Remove Duplicates, Trim + Clean, Replace non-breaking spaces (character 160) with a normal space, Capitalize Each Word on the name. How many rows and what is the name for C010?

Hint: Home > Remove Rows > Remove Blank Rows. Transform > Format > Trim / Clean / Capitalize Each Word. nbsp can be replaced with `Character.FromNumber(160)` in Replace Values (advanced editor) or by copying the character.

Session 18 - Power Query (Merge & Append)

Queries: Sales_Data, Products, Sales_Extra (load each as a connection-only query first).

S18.1 Merge Sales_Data with Products on ProductID using Left Outer. After expanding Category, UnitPrice, UnitCost: how many rows have null Category? How many rows would an Inner join give? How many for Left Anti?

Hint: Home > Merge Queries. Join Kind options. Nulls appear where the right table has no match.

S18.2 Does every product in Products appear in Sales_Data? Use a **Right Anti** merge (Products rows with no Sales match). How many rows?

Hint: Merge Products (first) with Sales_Data (second) and pick Left Anti, or Sales first and Right Anti.

S18.3 Append Sales_Data and Sales_Extra WITHOUT renaming anything first. How many rows and columns does the result have? How many nulls in Quantity and in Qty?

Hint: Home > Append Queries. Columns are matched by NAME, not position. Sales_Extra uses **Qty** and **Sales Rep**.

S18.4 Rename the mismatched columns in Sales_Extra (Qty -> Quantity, Sales Rep -> SalesRep) and append again. How many columns and rows now? Then Group By Region on the 2025 rows only: what are the four sums?

Hint: Rename in Sales_Extra BEFORE appending. Filter OrderDate >= 01-Jan-2025 after appending.

S18.5 Sales_Extra has blank Discount cells. How many nulls are there? Replace them with 0 (Replace Values or Transform > Replace Values null -> 0). What average discount (in %) do you get for Sales_Extra after replacement?

Hint: The Discount column shows 'null' in italics. Right-click header > Replace Values > Value To Find: null (type **null**).

S18.6 After the left-outer merge add a custom column `Profit = [SalesAmount] - [Quantity]*[UnitCost]`. What is total profit for the 794 rows that matched a product? (Handle nulls with `if [UnitCost] = null then null else ...`).

Hint: Sum ignores nulls. Use Transform > Statistics > Sum, or Group By.

Session 19 - Dashboard Design

Data: Sales_Data (+ Products helper columns). Goal: a one-page executive dashboard.

S19.1 Put these dashboard blocks in the recommended order top-to-bottom: Trends charts, Header/title, Insights/commentary, KPI cards. Also name the alignment/spacing rule you would follow.

Hint: Think: what does an executive read first?

S19.2 Build KPI cards for SalesAmount: **Total, Number of orders, Average order value, Min, Max**. What values should the cards show?

Hint: `=SUM(J2:J801)`, `=COUNTA(A2:A801)`, `=AVERAGE(J2:J801)`, `=MIN(...)`, `=MAX(...)`.

S19.3 Waterfall chart (valid products only, 794 rows): Gross revenue at list price (Quantity x UnitPrice) -> less Discounts -> Net revenue (SalesAmount) -> less Cost (Quantity x UnitCost) -> Profit. What are the values?

Hint: Gross = SUMPRODUCT of Quantity and UnitPrice (helper column). Discounts = Gross - Net. Insert > Waterfall; set Net and Profit as totals.

S19.4 Donut chart of Channel share and a **speedometer/gauge** for '% Delivered' with a target of 90%. What are the donut slices and what needle value does the gauge show?

Hint: Gauge = doughnut (or pie) with hidden half + a pie needle; needle = % delivered. Delivered % = `=COUNTIF(I2:I801,TRUE)/COUNTA(A2:A801)`.

S19.5 Critique: a dashboard uses 8 different bright colours, a 3-D pie of 10 slices and no titles. List three fixes.

Hint: Colour theory, chart choice and clarity.

Session 21 - Case Study: Mini Dashboard Project

Capstone using ALL techniques. Suggested flow: clean -> combine -> enrich (lookups) -> summarise (pivots) -> dashboard.

S21.1 Load & combine: with Power Query append Sales_Extra (after renaming mismatched columns) to Sales_Data. Confirm the row count, unique OrderIDs, and how many rows have a ProductID that does not exist in Products.

Hint: Remove Duplicates on OrderID and compare row counts; Left Anti merge to Products for orphans.

S21.2 Enrich: add Category, UnitCost and Profit (Profit blank for unknown products). What are combined Total Sales, Total Profit and the Profit margin (Profit / Sales of matched rows only)?

Hint: Merge with Products in Power Query, or use INDEX/MATCH or XLOOKUP helper columns.

S21.3 Year split: what are total sales for 2024 (Sales_Data) and for 2025 Jan-Mar (Sales_Extra)? Create a Year-Month column with TEXT/YEAR/MONTH to build the pivot.

Hint: Group dates by Years and Months or use `=TEXT(B2, "yyyy-mm")`.

S21.4 Insights: who is the #1 and #2 rep by total sales in the combined data, and what is the Profit by Category ranking (largest first)?

Hint: Pivot: SalesRep in Rows, Sales sum sorted desc. Category in Rows with Sum of Profit.

S21.5 Targets: using only 2024 actuals and the Targets sheet, in how many months did each region MISS its target (an INDEX/MATCH or pivot + conditional formatting approach)?

Hint: Same table as Session 12 - the 'Red (below)' column.

S21.6 People data: clean Employees names (Session 6), compute age (as of \$L\$2), grade (nested IF) and bonus eligibility (AND). What percentage of the 30 employees are bonus-eligible (Marks >= 75 AND Attendance >= 90)?

Hint: COUNTIF on the Eligible column / 30.

S21.7 Dashboard checklist: build a one-page dashboard with slicers for Region and Timeline (OrderDate). What values must the KPI cards show with no slicer selection (combined data, 850 rows)?

Hint: Card 1 = total sales, card 2 = orders, card 3 = average order value, card 4 = % delivered, card 5 = top rep. Add a line chart (monthly trend), a column chart (category), a donut (channel), and a text box of 3 insights.

PART B - Answer Key

Numbers come from the workbook exactly as delivered. If you change or sort the source data, re-check totals against a copy.

Session 1 - Interface, Layout & Data Types

S1.1 Numbers: **22**, text: **8**.

=COUNT(B2:B31) -> 22

=SUMPRODUCT(--ISTEXT(B2:B31)) -> 8

S1.2 As-is SUM = **56,524.00**; corrected total = **77,805.00**.

=SUMPRODUCT(--B2:B31) gives the corrected total without touching the data. Any of the four methods is fine as long as COUNT(B2:B31) becomes 30.

S1.3 Real dates: **20**, text dates: **10**. Earliest join date = **12-Jan-2022**.

Helper column: =IF(ISNUMBER(C2),C2,DATEVALUE(C2)), then =MIN(helper range).

S1.4 Active records = **16**.

=COUNTIF(D2:D31,TRUE)+COUNTIF(D2:D31,"Yes")

S1.5 Errors: **5**, text "N/A": **3**, average of valid scores = **72.23** (22 numeric scores).

=SUMPRODUCT(--ISERROR(E2:E31)), =COUNTIF(E2:E31,"N/A"), =AGGREGATE(1,6,E2:E31)

S1.6 Number = **1**, text = **2**, logical = **4**, error = **16**, array = 64.

S1.7 =C5*\$C\$2. Switch: File > Options > Formulas > tick **R1C1 reference style**.

Session 2 - Cell Referencing, Named Ranges, Data Entry Rules

S2.1 UnitPrice of P005 (Laptop-Dell) = **1,199**.

S2.2 =SUM(SalesAmt) = **2,110,571.95** (before adding the test row). After adding 1000 in J802 the total becomes 2,111,571.95. Delete the test row afterwards.

S2.3 The dropdown shows Mobile, Laptop, Tablet, Audio, Wearable. Typing 'Phone' is rejected with your message. (Named-range source is cleaner and works across sheets.)

S2.4 Rejected: **0** (below 1) and **12.5** (not whole). 12 and 50 are accepted.

S2.5 Rejected: **31-Dec-2023** and **01-Jan-2025**. (29-Feb-2024 exists because 2024 is a leap year.)

S2.6 =AND(LEFT(A2,4)="ORD-",COUNTIF(\$A\$2:\$A\$1000,A2)=1)

Test: typing ORD-1001 (already exists) or INV-5 is rejected; ORD-9999 is accepted.

S2.7 =\$A2*B\$1. E5 = quantity 20 x price of P004 (349) = **6,980**.

Session 3 - Sorting, Basic & Advanced Filtering

S3.1 Row 2 = **East, P001, 8,388.00**.

S3.2 **10** orders.

S3.3 10th largest SalesAmount = **10,439.55**.

S3.4 **34** rows.

S3.5 Blank rows = **3**; data rows remaining = **45** (this still includes duplicate customers - handled in Session 16B).

Session 4 - Advanced Filtering & Custom Views

S4.1 **42** rows.

S4.2 **4** rows.

S4.3 Reps: **Aisha Khan, Maria Lopez, Wei Chen** (3 names).

S4.4 **67** orders (Aisha Khan and Amit Verma).

S4.5 199 orders.

S4.6 View > Custom Views > choose the name > Show. Custom Views is greyed out when the workbook contains an **Excel Table** (Ctrl+T); convert the table to a range first. Always save an 'All' view first so you can get back.

Session 5 - Text Functions (Part 1)

S5.1 (a) 6 employees. (b) **169,414**.

=SUMPRODUCT(--(LEFT(C2:C31,2)="TX")), =SUMPRODUCT(--RIGHT(C2:C31,4))

S5.2 P010 = Tablet-Samsung: Category **Tablet**, Brand **Samsung**, brand length **7**.

S5.3 **Wearable-Samsung** with **16** characters. =MAX(LEN(B2:B21)) (enter as array in older Excel).

S5.4 Row 2 date is 02-Jan-2024 -> **Tuesday, 02 Jan 2024**.

S5.5 Row 3 SalesAmount = 5,988.00 -> ROUND(11.976) = **12** bars.

S5.6 **1800**; LEN = **8**.

S5.7 Code = WA-4531 -> **WA-31**.

Session 6 - Text Functions (Part 2: Cleaning)

S6.1 Raw shows as ' john miller ' -> **John Miller**.

S6.2 **28** names need cleaning; **32** characters removed.

S6.3 **5** codes.

S6.4 Raw (957) 557-1300 -> **9575571300**.

=SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(C2,"(", ""),"),", ""), " ", ""), "-",""), ".","")

S6.5 **15** rows.

S6.6 E008 = Sarah Wilson: first name **Sarah**, last name length **6**.

S6.7 Raw 759 Maple Avenue , Austin , TX 78735 -> **759 Maple Ave, Austin, TX 78735**.

Session 7 - Date & Time Functions

S7.1 E001 born 16-Dec-1988 -> age **37**. Employees over 40: **16**.

S7.2 Longest tenure = **11 years**: Peter Novak.

S7.3 E002: 26-Nov-2023 to 28-Nov-2025 -> **525** working days.

S7.4 Without holidays: 23. With holidays: **22** (26-Jan-2025 is a Sunday so only 01-Jan reduces the count).

S7.5 (a) **31-Jan-2024**, (b) **29-Feb-2024**.

S7.6 (a) **60**, (b) **260**, (c) **176**.

S7.7 DATE(2024,14,1) = **01-Feb-2025**; DATE(2024,3,0) = **29-Feb-2024**.

S7.8 Y = **1**, M = **21**, YM = **9**, MD = **16**.

TODAY() = date only, NOW() = date + time; both are volatile (recalculate on every change). Static date: **Ctrl+;** (date), **Ctrl+Shift+;** (time).

Session 8 - Logical Functions

S8.1

Grade	A	B	C	D	F
Employees	4	7	6	2	11

S8.2 E010 has 85 marks -> grade **B**.

=IFS(H2>=90,"A",H2>=75,"B",H2>=60,"C",H2>=50,"D",TRUE,"F")

S8.3 **5** employees.

S8.4 16 employees.

S8.5 11 employees.

S8.6

Tier	Platinum	Gold	Silver	Bronze
Orders	10	102	270	418

S8.7 11 orders.

Session 9 - Lookup Functions (VLOOKUP / HLOOKUP)

S9.1 Row 2 price = **349**. #N/A rows = **6** (ProductID P099 does not exist in Products).

S9.2 0 mismatches - SalesAmount = UnitPrice x Quantity x (1 - Discount) for every valid row; 6 rows show 'Missing product'.

S9.3 Total commission = **52,300.91**.

S9.4 They must match exactly: A/B/C/D/F counts from Session 8. If not, check the slab boundaries (90, 75, 60, 50, 0).

S9.5 Not directly (needs CHOOSE({1,2},...) trick). Answer = **P005**; use INDEX/MATCH or XLOOKUP instead.

S9.6 (a) **15** days, (b) **2** years.

S9.7 It silently returns the NEW column (or the wrong one) because the hard-coded 4 no longer points to UnitPrice. Fix: **MATCH("UnitPrice",Products!\$A\$1:\$F\$1,0)** as the column index, or use INDEX/MATCH / XLOOKUP which reference the return column directly.

Session 10 - Modern Lookup (XLOOKUP)

S10.1 6 rows.

S10.2 **P013** and **P018**.

S10.3 E002 has 4 bonuses. Last = **3,300** on 26-Aug-2025; first = **3,400** on 04-Feb-2025. E003 has no bonus -> **No bonus**.

S10.4 **3,990.00** (ORD-1777, 23-Dec-2024).

S10.5 4999.99 -> **2%**, 5000 -> **3.5%**, 10000 -> **5%**. match_mode = **-1**.

S10.6 One row that spills right: **Tablet-Samsung | Tablet | 549 | 370**.

S10.7 First: **1199** (Laptop-Dell). Second: **199** (Wearable-Fitbit, the first Wearable in the list).

Session 11 - INDEX-MATCH & MATCH-MATCH

S11.1 **299** (Tablet-Huawei).

S11.2 Highest: **Laptop-Dell (1,199)**; lowest: **Audio-Anker (59)**.

S11.3 (a) **45,500**, (b) **65,000**, (c) **66,500**.

S11.4 Target = **33,000**, actual = **33,760.20**, difference (actual - target) = **760.20**.

S11.5 (a) **20**, (b) **#N/A**, (c) **3** (7500 falls in the 5000 slab, the 3rd row).

S11.6 **=INDEX(Products!\$E\$2:\$E\$21,MATCH("P007",Products!\$A\$2:\$A\$21,0)) = 680** (Laptop-Lenovo). Inserted columns shift both ranges together.

Session 12 - Conditional Formatting

S12.1 **10** cells (= 2 x 5 duplicated customers). Each duplicate row is an exact copy of another row.

S12.2 Lowest = **59.00**, highest = **14,388.00**.

S12.3

Green	Yellow	Red
7	13	10

S12.4 10th value = **10,439.55**; cells highlighted = **10**. Bottom 5: 59.00, 59.00, 59.00, 59.00, 100.30.

S12.5

Region	Green (above)	Red (below)
East	5	7
West	4	8
North	6	6
South	6	6
Total	21	27

S12.6 =AND(\$I2=FALSE,\$J2>5000) applied to =\$A\$2:\$J\$801. **11** rows shaded.

Session 13 - Pivot Tables (Basics)

S13.1

Region	East	West	North	South	Grand Total
Sales	559,002.70	554,743.30	521,396.15	475,429.80	2,110,571.95

S13.2

Region	#1	Sales	#2	Sales	#3	Sales
East	Rahul Shah	264,136.70	Priya Patel	206,569.05	Amit Verma	88,296.95
West	Sneha Iyer	287,057.80	John Miller	169,512.45	Emily Clark	98,173.05
North	David Brown	212,635.80	Sarah Wilson	174,843.35	Carlos Ruiz	133,917.00
South	Maria Lopez	190,085.20	Wei Chen	186,438.15	Aisha Khan	98,906.45

S13.3

Category	East	West	North	South	Total
Mobile	165,265	156,840	120,588	134,427	577,120
Laptop	243,043	231,184	194,147	150,004	818,379
Tablet	52,660	71,979	72,830	85,428	282,896
Audio	49,883	50,370	68,003	55,658	223,915
Wearable	48,152	41,478	59,074	45,548	194,252
Unknown	0	2,893	6,753	4,365	14,011
Total	559,003	554,743	521,396	475,430	2,110,572

S13.4

Channel	Orders	Avg order	Total
Online	345	2,595.34	895,391.20
Retail	299	2,684.35	802,621.05
Partner	156	2,644.61	412,559.70
Total	800	2,638.21	2,110,571.95

S13.5 **89** orders, total **221,763.30**.

S13.6

Region	East	West	North	South
Share	26.5%	26.3%	24.7%	22.5%

Session 14 - Pivot Tables (Advanced)

Quarter	Q1	Q2	Q3	Q4
Sales	428,571.80	466,414.65	536,172.60	679,412.90

S14.1 Best month: **Nov** with **240,901.65**.

S14.2

Region	Q1	Q2	Q3	Q4
East	112,671	130,084	159,469	156,779
West	105,821	100,602	147,487	200,833
North	120,953	104,144	137,440	158,859
South	89,127	131,584	91,776	162,942

S14.3

Category	Sales	Profit	Margin
Mobile	577,120	175,415	30.4%
Laptop	818,379	241,849	29.6%
Tablet	282,896	85,336	30.2%
Audio	223,915	84,987	38.0%
Wearable	194,252	64,082	33.0%

S14.4 90 orders, **252,957.25** total sales.

S14.5 West = **220** rows; Grand Total = **800** rows.

S14.6

Rank	SalesRep	Sales
1	Sneha Iyer	287,057.80
2	Rahul Shah	264,136.70
3	David Brown	212,635.80

Session 15-16A - PivotCharts, Slicers, Timelines & Dashboards

S15-16A.1

Jan	Feb	Mar	Apr	May	Jun	Total H1
102,908	143,579	182,084	116,124	181,701	168,589	894,986
Jul	Aug	Sep	Oct	Nov	Dec	Total H2
158,899	158,866	218,407	215,024	240,902	223,487	1,215,586

S15-16A.2

Mobile	Laptop	Tablet	Audio	Wearable	Unknown
577,120	818,379	282,896	223,915	194,252	14,011
Online		Retail		Partner	
42.4%		38.0%		19.5%	

S15-16A.3 With 12 categories use a bar/column chart (sorted) - small slices in a pie cannot be compared.

S15-16A.4 Jul-Sep = **536,172.60**; Jul-Sep AND South = **91,776.35** (44 orders).

S15-16A.5 Report Connections lets one slicer filter several pivots. West total = **554,743.30** (220 orders).

S15-16A.6 (a) Line, (b) Column/Bar, (c) Pie/Donut, (d) Stacked column or area. Rules: limit to 3-4 colours, align objects to a grid, remove gridlines/clutter, label with units, use the same colour for the same measure everywhere.

Session 16B - Data Cleaning Tools & Error Handling

S16B.1 5 duplicate rows removed; 40 unique customers remain.

S16B.2 5 rows again. Emails uniquely identify a customer; names have inconsistent spacing/case so they miss duplicates. Best practice: clean first, then dedupe.

S16B.3 Distinct cities: 8 (Austin, Boston, Chicago, Dallas, Denver, Miami, Phoenix, Seattle). Customers in TX: 8.

S16B.4 Raw ' amit harris ' -> **Amit**.

S16B.5 31 replacements (duplicate rows included).

S16B.6 COUNT: 22 -> 30. Helper average = 58.85 (22 valid scores + 5 zeros = 27 values; 3 text 'N/A' ignored).

Session 17 - Power Query (Basics)

S17.1 OrderDate = **Date**, Discount = **Decimal number**, Delivered = **True/False**, SalesAmount = **Decimal (or Fixed decimal) number**, Quantity = Whole number. 800 rows.

S17.2 6 columns. Step name: Removed Other Columns (rename steps for clarity).

S17.3 711 rows.

S17.4 First order date 02-Jan-2024 -> 2024-01.

Region	East	West	North	South
Sales	559,002.70	554,743.30	521,396.15	475,429.80

S17.5 These must equal the pivot totals.

S17.6 Min 1001, max 1800.

S17.7 40 rows; C010 = **Amit Harris**.

Session 18 - Power Query (Merge & Append)

S18.1 Null Category rows = 6; Inner = 794 rows; Left Anti (rows in Sales with no product) = 6 rows.

S18.2 0 products never sold - every product sold at least once.

S18.3 850 rows and 12 columns (extra columns 'Qty' and 'Sales Rep'). Quantity: 50 nulls (all Extra rows); Qty: 800 nulls.

S18.4 850 rows, 10 columns.

Region	East	West	North	South
2025 Q1 sales	25,221.30	57,348.00	36,062.10	23,810.50

S18.5 7 nulls. Average discount after replacement = 3.80%.

S18.6 Total profit = 651,667.90 on sales of 2,096,560.90.

Session 19 - Dashboard Design

S19.1 Header -> KPI cards -> Trends -> Insights. Align to a grid (equal margins/gutters), keep white space, group related charts.

S19.2

Total	Orders	Average	Min	Max
2,110,571.95	800	2,638.21	59.00	14,388.00

S19.3

Gross	Discounts	Net revenue	Cost	Profit
2,179,767	-83,206	2,096,561	-1,444,893	651,668

S19.4

Online	Retail	Partner	% Delivered (needle)
42.4%	38.0%	19.5%	88.9% (target 90%)

S19.5 (1) Limit palette to 3-4 consistent colours, use accent only for exceptions (red/green vs target). (2) Replace 3-D pie with a sorted bar chart. (3) Add clear titles, units and KPI labels; align to a grid and remove clutter.

Session 21 - Case Study: Mini Dashboard Project

S21.1 850 rows, 850 unique OrderIDs, 6 rows with an unknown ProductID (all P099).

S21.2 Total sales (850 rows) = **2,253,013.85**; matched sales = 2,239,002.80; Total profit = **696,160.80**; margin = **31.1%**.

S21.3 2024 = **2,110,571.95**; 2025 Q1 = **142,441.90**.

S21.4 #1 **Sneha Iyer** (318,371.70), #2 **Rahul Shah** (277,189.40).

Category	Profit
Laptop	259,918
Mobile	185,121
Audio	92,837
Tablet	89,696
Wearable	68,588

S21.5

Region	Months below target
East	7
West	8
North	6
South	6

S21.6 5 of 30 = **16.7%**.

S21.7

Total sales	Orders	Avg order	% Delivered	Top rep
2,253,013.85	850	2,650.60	88.7%	Sneha Iyer